6. APPLICATIONS OF DIFFERENTIATION

Exercise 6.1

1. $AC = \frac{1}{10}x^2 - 4x - 20 + \frac{7}{x}$, $AVC = \frac{1}{10}x^2 - 4x - 20$, $AFC = \frac{7}{x}$,
$MC = \frac{3}{10}x^2 - 8x - 20$, $MAC = \frac{1}{5}x - 4 - \frac{7}{x^2}$

2. $C = ₹\ \frac{121}{6}$, $AC = ₹\ \frac{29}{12}$, $MC = ₹\ \frac{2}{3}$

3. $AC = x^2 - 2$, $MC = 3x^2 - 2$, $AR = 14 - x$, $MR = 14 - 2x$

4. $\eta_d = \frac{2}{x}$

5(i) $\eta_d = \frac{a - bx}{2bx}$, $x = \frac{a}{3b}$ (ii) $\eta_d = \frac{a - bx^2}{2bx^2}, x = \sqrt{\frac{a}{3b}}$

6. $\eta_s = \frac{4p^2}{2p^2 + 5}, \frac{36}{23}$ 7. $MR = \frac{50 - 2x}{5}$, 10, 0

8. $\eta_s = \frac{p}{2(p - b)}$, 1 11. $\eta_d = 4$

12. $P = -\frac{x^2}{100} + 160x - 120$, $AP = \frac{-x}{100} + 160 - \frac{120}{x}$, ₹ 147.90
$MP = \frac{-x}{50} + 160$, ₹159.80, $MAP = -\frac{1}{100} + \frac{120}{x^2}$, ₹ 1.19

13. x = – 8, 2 15. $n_d = \frac{p}{10 - p}, |n_d| = \frac{3}{2} > 1 \Rightarrow$ elastic.

16. P_E=₹ 30, x_E = 40 units 17. x = 2100 units, p = ₹ 130

18. x = 6 units

Exercise 6.2

1. AC is increasing when $x > 5$

3. P is maximum when $x \approx 46$, maximum profit ₹ 1107.68

4. Revenue is maximum when x = 220

5. Minimum value is –71, maximum value is 54

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